

Developing Organizational Agility through IT and Supply Chain Capability

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ABSTRACT

Organizations have increasingly invested money in information technology (IT) in order to improve their agility. It is generally believed that organizations with greater IT investment tend to be more agile to response to environmental changes. However, the issue of whether IT is an enabler or impeder of organizational agility still remains unresolved. Drawing upon resource-based view theory and the literatures of information systems and supply chain management, the authors develop and test a theoretical model that integrates IT capability, supply chain capability and organizational agility. The authors propose that IT capability enables the development of a higher level of supply chain capability which is embedded within inter-firm processes and in turn enhances organizational agility. Structural equation modelling is employed to test their theoretical conceptualization of 310 Australian fast-growth small-to-medium enterprises across different industrial sectors. The results show that IT capability does contribute to firm agility through enhancing inter-firm supply chain processes such as integration, information sharing and coordination. This research highlights the role of IT-enabled intermediated processes and the ways in which IT is used by firms to enhance core business processes.

Keywords: *Coordination, Information Sharing, Integration, Information Technology (IT) Capability, IT Resources, Organizational Agility, Resource-Based View of Firms, Supply Chain Capability*

INTRODUCTION

In the current context of intensive competition, globalization and time-to-market pressure, firms are making significant investments in information technology (IT) to develop agility and pursue fast and innovative initiatives so as to respond to environmental challenges. Agile

firms are able to deal with rapidly evolving situations, survive unexpected threats and thrive in competitive environments through capitalizing on emerging business opportunities (Lu & Ramamurthy, 2011). Therefore, agility is regarded as an imperative for business success, helping firms to achieve competitive performance in dynamic business environments (Fink and Neumann 2007; Nazir and Pinsonneault 2012; Sambamurthy et al., 2003).

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Research that investigates the relationship between IT and organizational agility is increasingly encountered in the information systems (IS) field. Some researchers (e.g., Nazir & Pinsonneault, 2012; Sambamurthy et al., 2003) assert that IT can enhance organizational agility by building digital options, helping firms to speed up decision making, facilitate communication, and respond quickly to changing conditions. Others (e.g., Van Oosterhout et al., 2006; Weill et al., 2002) argue that IT may hinder and even impede organizational agility because of inflexible legacy IT systems and rigid IT architectures. Ironically, a high level of IT investment may result in unintended “technology traps” over time (Grover & Malhotra, 1999, p. 907). In the digital business environment, although the increasing use of IT creates strong electronic linkages in supply chains, it may also have unintended adverse effects on supply chain flexibility and so may severely constrain supply chain performance (Gosain et al., 2004). For example, studies show that the integrated enterprise systems used to automate and support business processes have positive impacts on both business agility (Goodhue et al., 2009) and rigidity (Rettig, 2007). These mixed observations indicate that IT can be either an enabler or an impedier of organizational agility.

The use of IT in the supply chain context has also gained intensive attention in the IS area. While supply chains involve “the flows of material, information and finance among customers, suppliers, manufactures, and distributors” (Lee, 2000, p. 31), supply chain management is regarded as a digitally enabled inter-firm process capability (Rai et al., 2006). As IT provides new opportunities for firms to manage supply chain relationships, it is imperative that we understand how IT resources and capabilities relate to inter-firm business processes (Dong et al., 2009). Although research has examined the performance benefits of IT resources/capability (Bhatt & Grover, 2005; Stoel & Muhanna, 2009), there is still limited understanding of the links between IT capability and agility in the supply chain context (Kohli & Grover, 2008). Moreover, current literature

on IT business value has largely overlooked agility as a potential outcome, instead focusing on standard firm performance metrics (Nazir & Pinsonneault, 2012). Thus, further rigorous empirical examination is needed to understand how and why IT capability shapes firm agility through intermediate processes.

The present research attempts to address the above gaps in the literature. Drawing upon the resource-based view of the firm (RBV) theory and the IS and supply chain literature, we synthesize and theorize the commonly observed but understudied contradiction that relates to IT’s potential both to enable and to impede organizational agility. Recent IS research (Nazir & Pinsonneault, 2012; Tallon & Pinsonneault, 2011) suggests that the relationship between IT and firm agility depends on the extent of inter-firm processes. IT can increase firm agility through close integration and collaboration with supply chain partners, enabling organizations to sense and respond to market changes in an effective and efficient way (Dong et al., 2009; Rai et al., 2006). Following this logic, we expect that IT can help firms to gain agility through the development of IT capability and a higher level of IT-enabled supply chain capability which is embedded within inter-firm processes such as integration, information sharing, and coordination. For the purpose of the present study, IT capability is defined as a latent construct reflected in three dimensions: IT infrastructure, back-end integration, and IT human resources. We propose supply chain capability to consist of three interrelated processes: inter-firm integration, information sharing, and coordination and conceptualize market responsive agility as one type of organizational agility.

We examine the hypothesized linkages empirically based on data drawn from a survey of 310 fast-growth small-to-medium enterprises (SMEs) in Australia. We have chosen to test the proposed model using Australian SMEs because SMEs are a dominant part of and significant contributor to the Australian employment and economy (OECD, 2007). IS research on SMEs is still thin on the ground and the benefits SMEs derive from IT investments

are far from conclusive. Fast-growth SMEs are more entrepreneurial and risk taking in their business orientation. Focusing on fast-growth SMEs provides insights into how this cohort of firms leverages IT capability to develop their organizational capability in order to achieve business core competences.

This paper is structured as follows. The theoretical background section introduces the tenets of RBV, which forms the backbone of our conceptual model for hypothesis formulation. The research method section outlines the procedures used for data collection, validation of the measurement properties of the constructs, and the test of the proposed research model. Next we present our findings and finally conclude with a discussion of findings, implications for research and practice, limitations and potential avenues for future research.

THEORETICAL BACKGROUND AND HYPOTHESES

The RBV posits that the improvements of firm performance depend on availability of, or access to, valuable, rare, inimitable, non-substitutable and relatively immobile resources or resource bundles (Barney, 1991). According to the RBV, organizations succeed and achieve sustainable competitive advantage through treatment of resources/capabilities as central considerations in strategy formulation and as primary sources of competitive advantage. In the IS literature, the RBV has been used to explain how firms create business value from IT capability and organizational skills to leverage IT complementary resources (Bharadwaj, 2000; Wade & Hulland, 2004). Although IT resources (e.g., hardware and software) are rarely drawn upon for the purpose of creating and sustaining competitive advantage (Clemons & Row, 1991), IT capability helps organizations not only to create value but also to gain sustainable competitive advantage (Bharadwaj, 2000; Mata et al., 1995; Santhanam & Hartono, 2003).

According to Bharadwaj (2000), the combination of IT infrastructure, IT human resources,

and firms' ability to leverage IT for intangible benefits serve as firm-specific resources that lead to the creation of a firm-wide IT capability. Moreover, IS researchers (Santhanam & Hartono, 2003; Wade & Hulland, 2004) recommend the development of multidimensional measures of IT capability, derived systematically and theoretically, to allow the effective evaluation of the impact of IT capability on business performance. Predicated on these studies, this research defines IT capability as a firm's ability to acquire, deploy, combine and configure IT resources in order to support and enhance business strategies and processes (Wade & Hulland, 2004). We conceptualize IT capability as a multidimensional latent variable reflected in three elements: IT infrastructure, back-end integration and IT human resources. These three are critical resources that firms can utilize as they undertake their supply chain operations. IT infrastructure normally refers to physical IT assets including computers, communication facilities, shareable technical platforms and databases (Zhu 2004). In the supply chain context, back-end integration is regarded as a valuable IT resource for the digitally enabled supply chain which links web applications with back-office databases and facilitates supply chain operations between firms and their downstream and upstream partners (Zhu & Kraemer 2005). IT human resources are the skills and knowledge of a firm's IT personnel (Wade & Hulland, 2004). Although competitors can easily mimic a firm's IT resources, the way companies effectively combine IT resources within an organizational strategy so as to develop an overall IT capability is hard to acquire and difficult to imitate, thus providing firms with a source of competitive advantage.

Researchers (e.g., Wade & Hulland 2004, pp. 129-130) suggest that examining IT value creation should take into account "an indirect role for IT in firm performance. The basic logic is that IT affects other resources or processes which, in turn, lead to competitive advantage [...]. Therefore, researchers may find it particularly beneficial to use intermediate-level dependent variables at the business process,

department, or project level". In line with this view, the present research posits that IT capability can help firms to create value through the improvement of inter-firm processes in digitally enabled supply chains. The strategic management literature suggests that a high-level organizational capability that integrates and reconfigures resources, and fits with firm social, structural and cultural contexts, can be regarded as a source of performance (Grant, 1996).

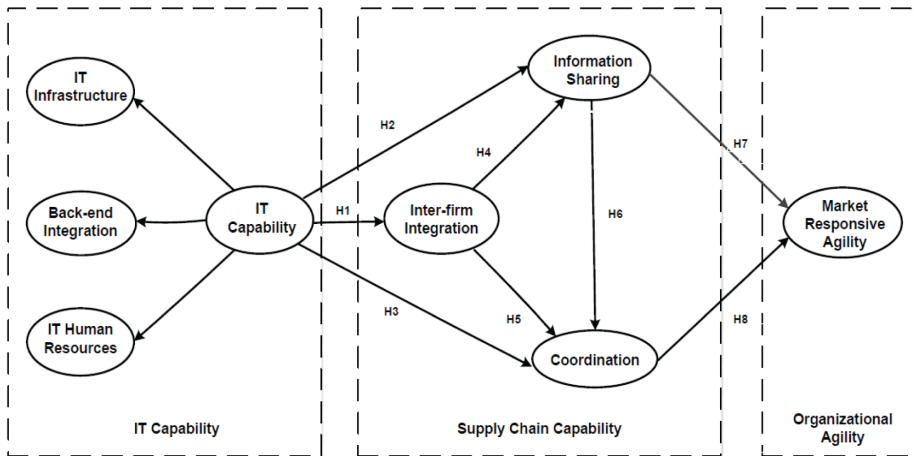
Prior IS research on the supply chain management field has focused on specific technologies and innovations such as electronic data interchange (EDI), vendor-managed inventory (VMI), and cellular manufacturing. However, researchers advocate that more investigations are needed to explore how IT capability helps firms to develop an inter-firm capability that can link firms with their supply chain partners so as to create business value (Dong et al., 2009; Kohli & Grover, 2008; Rai et al., 2006). Heeding this call, we define supply chain capability as a high-order IT-enabled organizational capability, which refers to a firm's ability to identify, utilize, and assimilate internal and external resources in order to enhance the entire supply chain activities (Wu et al., 2006). We further conceptualize supply chain capability as encompassing three dimensions: inter-firm integration, information sharing, and coordination, which represent typical yet important activities in the supply chain process (Lee et al., 2000). Each of these three dimensions reflects a firm's ability to undertake internal, cross-functional and inter-firm business activities within supply chains. Developing this kind of inter-firm capability is a long-term process which requires firms to make a series of strategically-integrated decisions and moves related to IT resources so as to blend them with organizational processes and knowledge resources (Barua et al., 2004). It is in this sense that IT-based inter-firm capability can be regarded as a valuable source of sustained competitive advantage.

According to Lu and Ramamurthy (2011), IT business value can be manifested in organizational agility, which helps firms to achieve

cost reduction, operational efficiency and sustainable competitive advantage. Organizational agility is a firm-wide capability to deal with and respond to unexpected environmental changes and respond to these changes by exploiting them as opportunities to grow and prosper (Overby et al., 2006). In the supply chain context, agility demonstrates a firm's ability to quickly respond to sudden and unexpected market changes through close collaboration with supply chain partners (Lee 2004). Lee (2004) opines that the best supply chains should not only be fast and cost-effective but also be agile, adaptable and aligned, highlighting that agility is critical for firms to survive and achieve long-term competitive advantage in fast-changing business environments. For example, Cisco uses e-hub technology to build strong digital connections with its manufacturers and partners. Such IT capability not only enables Cisco to take advantage of integrated supply chain processes, but also helps the company to enhance its agility when dealing with market demands, leading to new product development, market expansion and revenue growth. Following Lee (2004), we measure organizational agility from a market responsive perspective. We argue that firms able to employ IT capability to develop a high-level of IT-enabled supply chain capability which involves supply chain activity integration, real-time information sharing, and inter-firm coordination processes among supply chain partners are likely to develop their market responsive agility.

Based on the above discussion, the RBV and the perspective of organizational capability offer solid theoretical foundations explaining how and why firms with a high level of IT capability can achieve organizational agility through development of supply chain capability embedded with inter-firm processes in their supply chains. Firms achieve competitive advantage not only from commonly available IT resources but also from the integration of these IT resources to form a valuable IT capability which can then be leveraged to develop a higher-order, IT-enabled organizational capability

Figure 1. Hypothesized model



residing in organizational skills and processes rather than in IT assets (Bharadwaj, 2000; Rai et al., 2006). Figure 1 depicts a hypothesized model of IT capability, supply chain capability and organizational agility, and is followed by a discussion and formulation of testable hypotheses.

As already noted, IT capability reflects three dimensions: IT infrastructure, back-end integration, and IT human resources. These three dimensions complement each other, enabling firms to develop a higher level of organizational capability: supply chain capability which includes inter-firm integration, information sharing, and coordination. IT infrastructure refers to physical IT assets such as computers, communication facilities, shareable technical platforms and databases, which provide a solid platform upon which firms can leverage technologies not only to conduct business activities but also develop a flexible technology structure (e.g., integrated database) in order to respond to customer demands and market changes for business development (Zhu, 2004). A solid IT infrastructure can foster strong links between firms and their supply chain partners, leading to high levels of integration, information sharing, and coordination in supply chains (Bi et al., 2010; Zhu & Kraemer, 2005). In addition, back-end integration, as an intangible IT resource, drives

collaborative connections among supply chain partners and enhances the flow of information among supply chain partners (Zhu & Kramer, 2005), adding value to integration (Rai et al., 2006), collaborative planning, forecasting, and replenishment (Bi et al., 2011) and transactions among supply chain partners (Dong et al., 2009). Finally, IT human resources complement IT physical assets, provide knowledge and skills to develop appropriate IT applications so as to support business strategies and improve inter-firm supply chain processes, helping firms to conduct supply chain activities effectively and efficiently (Bi et al., 2010; Fink & Neumann, 2007). Therefore, a firm with superior IT capability involving IT infrastructure, back-end integration and IT human resources are able to enhance the overall supply chain capability through closer integration of decisions and operations, timely information sharing and effective supply chain coordination activities (Rai et al., 2006). Thus, we hypothesize that:

- H1:** IT capability is related positively to inter-firm integration.
- H2:** IT capability is related positively to information sharing.
- H3:** IT capability is related positively to coordination.

Inter-firm integration is the extent to which firms collaborate on strategic planning and forecasting activities with their supply chain partners (Wu et al., 2006). In the context of supply chain operations, a firm's ability to effectively integrate strategic supply chain activities with partners is a prerequisite to achieving high level of supply chain information sharing and coordination efficiency (Cao & Zhang, 2011). Inter-firm integration not only facilitates joint production planning and sales forecasting (Rai et al., 2006), joint resource planning and work scheduling (Kim et al., 2006), but also enhances joint process integration among members (Johnson et al., 2007). Studies show that firms employing strategic integration with supply chain partners are likely to improve coordination and information exchange activities (Stank et al., 2001), and to increase the overall efficiency of production or exchange through closer integration of decisions and operations (Dong et al., 2009). Thus, we hypothesize that:

H4: Inter-firm integration is related positively to information sharing.

H5: Inter-firm integration is related positively to coordination.

While information sharing refers to the effective and efficient exchange of knowledge between firms and supply chain partners, coordination refers to firms' ability to coordinate transactional related activities with their partners (Wu et al., 2006). A typical supply chain network involves collecting, interpreting, storing, and sharing data through effective information exchange between members in order to improve efficiency in coordination activities (Lee et al., 2000). Effective information sharing among supply chain members leads to supply chain capability by increasing coordination, flexibility, and responsiveness (Lee 2000). Kim et al. (2006) suggest that supply chain partners exchanging information with each other in a frequent and time manner can contribute to inter-firm coordination. Thus, we hypothesize that:

H6: Information sharing is related positively to coordination.

Market responsive agility, as one type of organizational agility, involves knowledge management, through which appropriate responses to environmental changes or new market development can be identified (Kim et al., 2006). Market responsive agility includes the scanning and processing of extensive amounts of information to identify and anticipate external changes, and also involves continuously monitoring and quickly improving product/service offerings in response to market and customer needs (Lu & Ramamurthy, 2011). In contemporary volatile marketplaces, it is imperative for firms to develop a responsive agility so as to constantly collect, monitor and process changing environmental signals, make innovative decisions, and quickly adjust processes to capitalize on market opportunities, thus facilitating the achievement of sustainable competitive advantage (Sambamurthy et al., 2003). Kim et al. (2006) argue that effective and efficient inter-firm processes in supply chains can help firms to accommodate market changes or customer requests in a timely manner through efficient information exchange and coordination activities. Thus, we hypothesize that:

H7: Information sharing is related positively to market responsive agility.

H8: Coordination is related positively to market responsive agility.

RESEARCH METHODOLOGY

Target Population and Survey Sample

The data used for testing our hypothesized model was collected through an online survey of 1,335 Australian fast-growth SMEs compiled by Business Review Weekly (BRW). The BRW Fast Growth enterprises are similar to Fortune's FSB 100 annual list of North America's fastest

growing small companies. Key inclusion criteria for SMEs to enter the BRW fast-growth project are that their previous year's turnover must exceed AUD\$500,000; they must have fewer than 200 full-time employees; they cannot be a subsidiary of an Australian or overseas corporation; and they must not receive more than 50% of their revenue from a single client. Except for the turnover criterion, which is subject to indexing, the other criteria have remained constant. Fast-growth companies from this sample fall within Ghobadian and O'Regan's (2000) definition of SMEs.

Data Collection Procedures

A personalized email highlighting the academic nature of the study was sent to either the founder or CEO of all 1,335 fast-growth SMEs. In our emails, we emphasized the importance of having respondents with a good understanding and

overview of their firm's e-business activities to participate in our survey, urging the founder or CEO to personally complete the online questionnaire, where possible. A follow-up email was sent three weeks after the initial one, and a second reminder email another two weeks later. Respondents were assured of confidentiality. A total of 310 responses were obtained, which gave a gross response rate of 28.1%, after discounting 195 incorrect email addresses and 32 SMEs which declined to participate. All responses were filled by either the company founder or its CEO.

We first tested the sample for non-response bias, using the approach suggested by Armstrong and Overton (1977). Differences in responses to all the constructs between early respondents (i.e., those that completed the survey upon the first invitation) and late respondents (i.e., those who replied to follow-up emails) were compared. Independent sample *t*-tests

Table 1. Profile of responding firms

Demographic	% (<i>n</i> =310)
Industry	
Information Technology	18.8
Property & Business Services	18.1
Personal & Other Services	9.6
Finance & Insurance	8.9
Communications	6.6
Other ^a	38
Company Age	
Less than 5 years	49
More than 5 years	51
Previous Year Growth Rate	21.9-759.5
CEO/Founder's Education Level	
Tertiary	53.9
MBA	16.6
Year 12	13.7
PhD or Doctorate	1.8
Other	14.0

Note. ^a Other industry sectors include Construction, Retail Trade, Manufacturing, Health & Community services, Wholesale Trade, Education, Transport & Storage, Accommodation, café, restaurants, Mining, Cultural & recreational services.

on each construct failed to reveal significant differences between early and late respondents (all p -values $> .05$), suggesting that non-response bias was not an issue.

The profile of the responding firms in our study (Table 1) shows that they represent all major industry sectors. There is also equal distribution of companies in terms of their age (or years of establishment). All responding firms had achieved a growth rate in excess of 20%.

Common Methods Bias

As our study used a self-administered questionnaire and respondents were in a senior management position qualified to assess firm performance, measurement was subject to cognitive biases due to participants "seeking to present themselves in a favorable manner" (Thompson & Phua, 2005, p. 541). Anticipating such a possibility, we incorporated Marlowe and Crowne's (1961) Social Desirability Scale in our online questionnaire, inviting participants to complete this section as part of the survey. The incorporation of Marlowe and Crowne's (1961) Social Desirability Scale enabled us to assess all study items for social desirability response bias in order to address internal validity and psychometric aspects of instruments. Marlowe and Crowne's (1961) Social Desirability Scale has been used widely for checking cognitive biases (Ballard 1992). In this study, we tested common methods bias using structural equation modeling (SEM) procedures recommended by Podsakoff et al. (2003) to examine the influence of social desirability on the research constructs. We found no significant relationships between the social desirability construct and the research constructs (all p -values $> .05$). Accordingly, social desirability does not contribute significantly to the model, suggesting that there is no common method bias.

Constructs

Measurement items were developed based on a comprehensive review of the literature (Table 2). Development of respective measurement models incorporate successive stages of theoretical modeling, statistical testing, and refinement (Straub, 1989).

Instrument Validation

Data were analyzed with AMOS 17.0, using confirmatory factor analysis (CFA) procedures with the maximum likelihood (ML) estimation method. Prior to conducting the CFA, we ran an exploratory factor analysis (EFA) on all indicators. Principal axis factoring with direct oblimin rotation yielded consistent groupings with our hypothesized measurement models. All constructs were tested for reliability, validity, and fit. Based on an assessment of CFA fit statistics, measurement models were further refined to obtain sound fit. Respectively, Tables 3 and 4 show correlations and descriptive statistics and measurement properties of constructs. As reported below, instrument validation proceeded through four steps: calculation of construct reliability; variance extracted estimates; and evaluation of convergent and discriminant validity.

Construct Reliability

Construct reliability, a measure of consistency, assesses the degree to which items are free from random error. Indicator and composite reliability are two measures of construct reliability (Fornell & Larcker, 1981). While indicator reliability represents the proportion of variation that is explained by a construct it purports to measure, composite reliability reflects the internal consistency of indicators (Werts et al., 1974). In the present study, indicator reliability values range between .43 and .95, and composite reliability values exceed the recommended value of .70 (Nunnally & Bernstein, 1994).

Table 2. Constructs and indicators

Constructs	Indicators
1. IT Infrastructure (ITIF) <i>Adapted from Zhu (2004)</i>	Our company has a good telecommunication infrastructure.
	Our company's IT systems infrastructure is very flexible in relation to company's future needs.
	Our company's IT systems enable us to effectively cooperate electronically with suppliers/partners and customers.
2. Back-end Integration (BI) <i>Adapted from Zhu and Kraemer (2005)</i>	There are well-integrated multiple web applications encompassing different areas in our company.
	Our company shares common databases for various applications, rather than having a separate database for each application.
	Our company's databases are electronically integrated with our supply chain partners.
3. IT Human Resources (ITHR) <i>Adapted from Bharadwaj (2000)</i>	Our company hires highly specialized or knowledgeable people for e-business.
	IT people working for our company are generally aware of functions of e-business.
	IT people working for our company are adequately trained in e-business.
4. Inter-firm Integration (INTE) <i>Adapted from Kim et al. (2006)</i>	Our supply chain has built-in functions to collaborate on forecasting and planning with our supply chain partners.
	Our company projects and plans future demand collaboratively with our business partners through supply chain.
	Our supply chain allows us to project and plan future demand collaboratively with our business partners.
	Collaboration in demand forecasting and planning with our business partners is something we always do through our supply chain.
5. Information Sharing (INFS) <i>Adapted from Kim et al. (2006) and Wu et al. (2006)</i>	Our company exchange more information with our supply chain partners than our competitors do with theirs.
	Information flows more freely between our company and supply chain partners than between our competitors and theirs.
	Our information sharing with supply chain partners is superior to the information shared by our competitors from theirs.
6. Coordination (COOR) <i>Adapted from Kim et al. (2006) and Wu et al. (2006)</i>	Our company conducts transaction follow-up activities more efficiently with our supply chain partners than do our competitors with theirs.
	Our company spends less time on supply chain coordination transactions with our supply chain partners than our competitors with theirs.
	Our company conducts supply chain coordination transactions at less cost than do our competitors with theirs.
7. Market Responsive Agility (MRPA) <i>Adapted from Kim et al. (2006) and Wu et al. (2006)</i>	Compared with our competitors, our company responds more quickly and effectively to changing customer and supplier needs.
	Compared with our competitors, our company responds faster and more effectively to changing competitor strategies.
	Compared with our competitors, our company develops and markets new products more quickly and effectively.
	Compared with our competitors, our company is competing effectively in most markets.

Table 3. Correlation matrix, mean scores and standardized deviations

	Mean	SD	1	2	3	4	5	6	7
1. ITIF	5.53	1.08	.81						
2. BI	4.12	1.63	.39**	.71					
3. ITHR	4.95	1.69	.48**	.52**	.83				
4. INTE	4.30	1.60	.27**	.35**	.30**	.91			
5. INFS	4.36	1.36	.35**	.35**	.31**	.44**	.92		
6. COOR	4.40	1.24	.38**	.34**	.31**	.47**	.71**	.88	
7. MRPA	5.35	1.07	.29**	.32**	.34**	.38**	.41**	.51**	.75

Note. (1) * $p < .05$. ** $p < .01$. (2) The diagonal elements are the square root of the AVE.

Table 4. Confirmatory factor analysis: Standardized loadings and reliability

Constructs	Cronbach's α	Construct Reliability	Variance Extraction	Range of Standardized Loadings	Range of Indicator Reliability
1. ITIF	.83	.89	.66	.74 - .88	.55 - .78
2. BI	.75	.75	.50	.66 - .79	.43 - .62
3. ITHR	.86	.87	.69	.75 - .95	.56 - .91
4. INTE	.95	.95	.83	.85 - .95	.71 - .89
5. INFS	.94	.95	.85	.89 - .91	.79 - .95
6. COOR	.91	.91	.77	.85 - .91	.72 - .83
7. MRPA	.83	.84	.56	.67 - .80	.45 - .64

Note. All factor loadings are significant at $p < .001$ level

Variance Extracted Estimate

Variance extracted estimate reflects the overall amount of variance in indicators accounted for by a latent construct (Fornell & Larcker, 1981). In this study, all estimates exceed the recommended value of .50 (Hair et al., 2006).

Construct Validity

Construct validity was established by measuring convergent and discriminant validity of measurement items (Phillips and Bagozzi 1986). Convergent validity assesses the consistency across multiple operationalizations. Values for t -statistics for all factor loadings were found to be significant (all p -values $< .001$), indicating that measures satisfy convergent validity criteria

(Gefen et al., 2000). According to Fornell and Larcker (1981), average variance extracted for each construct should be greater than the squared correlation between constructs when assessing discriminant validity, the extent to which different constructs diverge from one another. In this case, results suggest that items share more common variance with related than non-related constructs, with all constructs meeting this criterion.

Data Analysis

Confirmatory and full structural model fit were assessed using multiple indices (Hair et al., 2006), including the normed chi-square (χ^2/df), comparative fit index (CFI), Tucker-Lewis Index (TLI), root mean-square error of

approximation (RMSEA), and standardized root mean-square residual (SRMR). All seven measurement models tested were found to meet the criteria set for these indices (Hair et al., 2006): χ^2/df ratio < 3; CFI and TLI > .90; RMSEA < .08; and SRMR < .08.

IT capability was modeled as a reflective second-order construct comprised of three first-order dimensions: IT infrastructure, back-end integration and IT human resources. According to Jarvis et al. (2003), the first-order factors are complementary (i.e., they interact and co-vary with each other, the covariance of these three first-order factors ranges between .49 and .62). A reflective second-order construct is appropriate for capturing complementarities (Tanriverdi & Venkatraman, 2005). The alternative approach of using a formative second-order modeling is not appropriate because it does not assume any interactions or covariance among the first order dimensions of a higher-order construct (Chin, 1998).

Data fit the measurement model for IT capability well: $\chi^2(24)=43.802$, $\chi^2/\text{df}=1.825$, CFI=.984, TLI=.977, SRMR=.05, RMSEA=.04. Cronbach's α , construct reliability, and variance

extraction for IT capability are $\alpha=.85$, CR=.79, and VE=.57 respectively. As theorized in the Theoretical Background and Hypotheses section, IT capability is a reflective higher-order construct comprising multiple dimensions with significant loadings (all p -values<.001). Paths from second-order construct to first-order factors are of high magnitude, either nearing or exceeding a suggested cutoff value of .7 (Chin, 1998). Marsh and Hocevar (1985) suggested that the efficacy of second-order models should be assessed by the target coefficient (T ratio) with an upper bound of 1. Our models display very high T ratios approximating 1, implying that relationships among first-order constructs are sufficiently captured by their respective second-order construct (Stewart & Segars, 2002). Given solid theoretical and empirical grounds, and the parsimonious nature of the second-order factors (Hull et al., 1991), the conceptualization of IT capability as a reflective, high-order, multidimensional construct is considered justified.

Table 5. Proposed hypotheses and test results

Hypothesis	Standardized Path Estimates	Conclusion
H1. IT Capability → Inter-firm Integration	.44***	Supported
H2. IT Capability → Information Sharing	.37***	Supported
H3. IT Capability → Coordination	.27**	Supported
H4. Inter-firm Integration → Information Sharing	.27***	Supported
H5. Inter-firm Integration → Coordination	.16**	Supported
H6. Information Sharing → Coordination	.59***	Supported
H7. Information Sharing → Market Responsive Agility	.05	Not Supported
H8. Coordination → Market Responsive Agility	.55***	Supported
Model Fit Indices		
$\chi^2(219)=404.271$ $\chi^2/\text{df}=1.846$ CFI=.964, TLI=.959 SRMR=.057 RMSEA=.052		

Note. * p <.05. ** p <.01. *** p <.001.

RESULTS

Given the acceptable measurement models, we estimated a full latent variable structural model (Anderson & Gerbing, 1988) using same goodness of fit criteria to test our structural model and respective hypotheses. Table 5 summarizes the results of hypotheses testing, revealing a reliable and robust fit between our theoretical model and sample covariances: $\chi^2(220)=404.617$, $\chi^2/\text{df}=1.839$, CFI=.964, TLI=.959, SRMR=.058, RMSEA=.052. These indices suggest a good model fit. The squared multiple correlation (SMC) values, which are similar to R^2 in regression analysis, show that this model accounts for 19% of the variance in inter-firm integration, 30% of the variance in information sharing, 61% of the variance in coordination, and 35% of the variance in market responsive agility. Table 5 shows that all hypothesized relationships, except H7, are supported.

We adopted the three-step method suggested by Baron and Kenny (1986) to test the mediating effects of supply chain capability. As Table 6 shows, the direct links between IT capability and market responsive agility is partially mediated by the supply chain capability, which includes supply chain integration, supply chain information sharing and supply chain coordination.

DISCUSSION

Aiming to address the controversial and understudied issues about IT and organizational agility, this study explores the role of IT resources/capability in developing a higher level of IT-enabled supply chain capability and how this supply chain capability helps Australian fast-growth SMEs to enhance market responsive agility. Our results show that seven of eight hypothesized relationships are supported, the exception being the link from information sharing to market responsive agility. Findings suggest that IT capability has a substantial effect on development of a higher level of IT-enabled supply chain capability in the SME context. This IT-enabled capability is deeply embedded into inter-firm supply chain processes such as integration, information sharing, and coordination among members. The development of such process integration, collaboration and coordination capability that leverages IT resources requires significant time and is specific to a firm's social, structural, and cultural context. This renders it hard to be imitated by competitors and thus underlines why it is a source of performance.

The findings also highlight that such effective and efficient inter-firm supply chain processes can help fast-growth SMEs to respond to market changes or customer requests in a timely manner through efficient information exchange and coordination activities. This then contributes to organizational agility which

Table 6. Results of mediating tests

Standardized Path Estimates							
IV	M	DV	IV→DV	IV→M	IV+M→DV		Mediating
					IV→DV	M→DV	
IT Capability	COOR	MRPA	.50***	.52***	.30***	.42***	Partial
IT Capability	COORD+INFS	MRPA	.50***	.37***	.27***	.41***	Partial
IT Capability	COORD+INFS+INTE	MRPA	.50***	.16***	.30***	.42***	Partial

Note. * $p<.05$. ** $p<.01$. *** $p<.001$.

is critical to effective competition in rapidly fast changing environments. Regarding the link between information sharing and market responsive agility, a possible explanation is that the direct effects of information sharing on market responsive agility are mediated through the coordination because of positive sequential links between information, coordination, and market responsive agility. This means that the impact of supply chain capability on agility is a causally complex inter-firm process that moves from integration to information sharing and finally to coordination which is related positively to market responsive agility.

Compared with larger companies, SMEs in general lack the resources to implement extensive IT innovations. However, there is growing evidence suggesting that SMEs are able to achieve IS success by employing IT resources and developing IT competence. For example, Caldeira and Ward (2003) indicated that IT human resources are imperative for SMEs to gain business advantage. Eikebrokk and Olsen (2007) suggested a positive relationship between IT competence and IS success in European SMEs. Similarly, Johnston, Wade, and McClean (2007) surveyed 1666 SMEs in North America and Europe, finding that successful adoption and use of IT leads to financial benefits for SMEs in all nations studied. More recently, Cragg, Caldeira, and Ward (2011) highlighted that organizational IS competence in SMEs depends on how IT resources are leveraged in support of core competencies at the firm level. Our findings lend support to these views.

This study contributes to extant theory, research, and practice in six salient ways. First, a fundamental contribution relates to developing, theorizing, and empirically validating a theoretical model investigating nomological relationships among IT resources/capability, supply chain capability, and organizational agility. This research empirically tests the application of RBV theory, thus offering a sharp theoretical lens to view the phenomena in IT business value creation.

Second, this study provides initial empirical evidence from an investigation of essential IT

capability and its relationship with organizational agility. Antithetical findings concerning IT-agility are not only uncommon but research in this area is also relatively sparse (Lu & Ramamurthy, 2011, Nazir & Pinsonneault, 2012). Therefore, understanding whether IT enables or impedes agility still remains unclear. The present research addresses this issue, suggesting that IT capability can help firms to enhance organizational agility by integrating inter-firm business processes, with a rigorous examination of the interrelationships between IT capability, IT-enabled supply chain capability, and organizational agility.

Third, this study bridges insights from the IS and supply chain management literatures to examine the role of IT in building supply chain capability and its consequence on firm agility. Understanding how IT impacts supply chain management is important for firms to achieve business advantage in dynamic business environments. We highlight IT as a critical enabler in building agile and adaptable supply chain capability which involves strategic integration, timely information sharing, and effective coordination processes among members. Because this kind of IT-based organizational capability is firm specific and hard for competitors to imitate, it is regarded as a source of competitive advantage.

Fourth, we conceptualize IT capability as a higher level and multidimensional construct reflecting commonality among three dimensions: IT infrastructure, back-end integration, and IT human resources. IT infrastructure and back-end integration represent IT physical assets providing platforms for firms not only to conduct business activities but also to facilitate integration, information sharing, and coordination processes among supply chain partners. IT human resources offer knowledge and skills to develop appropriate IT applications in order to support business strategies and inter-firm supply chain processes. This conceptualization emphasizes the complementarity among IT resources that provide firms with a robust, stable, and efficient foundation for organizational agility. The theme of IT resource complementarity

also highlights that it is imperative for firms to simultaneously develop an adequate competency level in these three elements in order to successfully manage IT, enhance agility and thus realize value.

Fifth, this study contributes to IS research by investigating the business value of IT in the SME context. It is noteworthy, however, that compared with their larger counterparts, SMEs gain little attention and research targeting fast-growth enterprises is almost non-existent in the IS field (Bi et al., 2010). Fast-growth SMEs are entrepreneurially-oriented, “willing to take risks, and tend to initiate aggressive competitive actions” (Upton et al., 2001, p.61). Fast-growth enterprises foster corporate cultures that capitalize on turbulence and any apparent chaos, seek competitive opportunities to add profitable value to their products or services, and share relevant customer- or market-focused information with key stakeholders (Barringer et al., 2005). Our findings highlight that it is essential for companies to adopt a proactive IT stance, experiment and explore new technologies in order to exploit existing competencies, address and create new business opportunities.

Lastly, this study contributes to practice by providing a theoretical framework for managers to understand the way in which IT helps firms achieve organizational agility. Although many companies invest large amounts of money in IT and are increasingly aware of agility in light of the rate of change in global markets, there is limited understanding among executives as to how IT is related to firm agility. Our research demonstrates the synergistic role of different IT resources in building IT capability, which enhances firm agility through the development of higher-order supply chain capability, helping managers to understand the interrelationships between IT resources/capability, supply chain capability, and firm agility.

This study has three important implications for management. First, this study highlights that resources/capabilities become sources of competitive advantage only when they are exploited through business processes. Managers should assign a high priority to the identifica-

tion of resource competencies that have strong potential for developing specific organizational capabilities and focus on appropriate business processes where these capabilities are deployed.

Second, we conceptualize IT capability as a multidimensional construct which captures the commonality among IT physical assets and IT human resources. This conceptualization emphasizes the complementarity among IT resources that together enhance inter-firm supply chain process capabilities. Managers should understand the complementary role of IT resources in representing IT capability and therefore need to develop these three dimensions simultaneously at an appropriate level in order to successfully manage IT and thus facilitate the inter-firm supply chain processes.

Third, we show that inter-firm supply chain processes exert a significant impact on organizational agility, which firms must achieve if they are to sustain business success in dynamic and volatile environments. Due to the interconnectivity enabled by digital technology on a global scale, firms are no longer working alone and thus the competition will not only be between companies, but also between supply chains. Firms like Dell, Cisco and Wal-Mart gain substantial benefits from the establishment of integrated, inter-firm processes in their supply chains. Managers should bear in mind that building strategic integration, collaboration, and coordination with supply chain partners is critical when doing business, particularly in dynamic environments.

LIMITATIONS AND FUTURE RESEARCH

This study has a number of methodological and conceptual limitations. First, IT capability- and IT-enabled supply chain capability-building processes and realization of agility have dynamic features that evolve over an extended period of time. The present research adopts a static cross-sectional research design with data collected at a single point in time. This approach is limited in addressing processes-oriented is-

sues or causal relationships. Future research might consider using longitudinal designs to address themes relating to the causal dynamics of capability-building processes and the ongoing processes between capability and agility.

Second, utilizing a single-informant (CEO and/or founder) data collection technique presents problems of data credibility. Single informant studies are well-known for their susceptibility to reporting bias. Future research might consider obtaining data from managers across the IT, production and operations functions.

A third limitation relates to sample characteristics upon which the present hypotheses are tested. The current investigation is drawn from a relatively small proportion of self-selected fast-growth SMEs in a specific western region. While the present hypothesized model might be applicable to larger firms as well as SMEs in other Asian locales such as Chinese companies, further research is needed to extend generalizability of the findings.

Finally, the present study only explores the utilization of IT capability in enhancing organizational agility. Technology is only one piece of the puzzle in achieving agility from a socio-technical perspective (Bostrom & Heinen, 1977). Future research should extend the current investigation and explore other elements such as how culture, structure and leadership interact with IT in enabling agility. Including these variables could offer significant improvements over the current model, providing a more comprehensive understanding of the value of IT complementary resources and IT capability.

CONCLUSION

Due to tremendous environmental changes and uncertainties, organizational agility is regarded as a key competitive imperative. However findings related to the relationship between IT and organizational agility is far from conclusion. While a few studies suggest the enabling role of IT on agility, others argue that IT impedes agility. We sought to better understand this

commonly observed but understudied IT-agility contradiction. We refined the conceptualization and measurement of IT capability as a latent construct reflected in its three dimensions: IT infrastructure, back-end integration and IT human resources. We argue that IT capability enables firms to gain organizational agility through the development of a higher-level of IT-based supply chain capability. The current findings demonstrate that fast-growth firms foster, nurture and develop firm-wide IT capability by successfully managing and leveraging their IT resources in such a way as to make these enterprises agile. In the light of the ongoing debate about the business value of IT, we hope this study motivates further discussion and encourages the advancement of theory that helps us to improve our understanding of the dynamics of IT and its relationship to organizational agility.

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